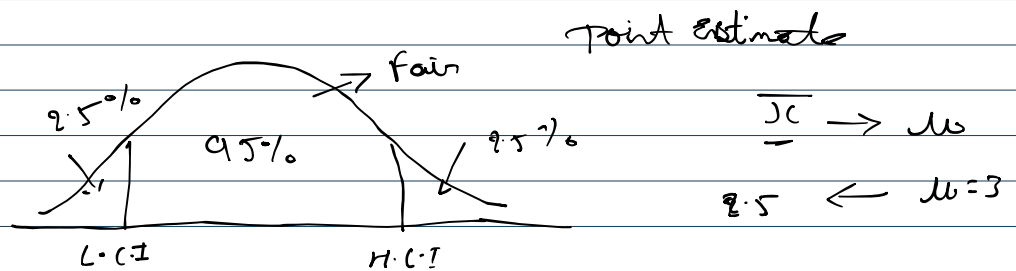


Confidence Interval and Margin of Error



Confidence Interval :- point estimate $\pm$ Margin of Error

$$\begin{aligned}\text{Margin of Error} &= z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \\ &= \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\end{aligned}$$

Eg:- on a section of exam, the standard deviation is known as 100. A sample of 30 test takers has a mean of 520. Construct a 95% C.I about the mean.

$$\sigma = 100 \quad n = 30 \quad \alpha = 0.05 \quad \bar{x} = 520$$



$$\begin{aligned}\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \\ 0.05/2 = 0.025\end{aligned}$$

$$\begin{aligned}\text{z-score} \\ \underline{+1.96} \quad z_{\alpha/2} \quad 0.05\end{aligned}$$

$$\begin{aligned}\text{Lower C.I} &= \bar{x} - (1.96) \frac{100}{\sqrt{30}} \\ &= 520 - (1.96) \frac{100}{\sqrt{30}}\end{aligned}$$

$$\text{Lower C.I} = \underline{480.8}$$

$$\text{higher Confidence Interval} = 520 + (1.96) \frac{100}{\sqrt{30}}$$

$$= 559.2$$

$$\begin{aligned}\text{L.C.I} \quad \text{H.C.I} \\ \underline{(480.8, 559.2)}\end{aligned}$$

$$\begin{array}{ccc} \text{L.C.I.} & & \text{U.C.I.} \\ \text{||} & & \text{||} \\ (480.8, 559.2) & & \\ \text{└───┐} & & \\ & 1 & \\ & 520 & \end{array}$$

Conclusion $\rightarrow$ I am 95% confident that CAT score has
 as Avg $\neq 520$